

MH1201 Linear Algebra II

Final Exam (Solutions)

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Overview

- Topics: matrix-space linear maps, diagonalisation with repeated eigenvalues, polynomial inner products, all-ones matrices, and existence of linear maps under dimension constraints.
- The pack follows the reference solution style: each question has a problem statement, solution methods, and a detailed marking scheme.
- Alternative methods are included only where they add a useful different viewpoint.

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Question 1**(20 Marks)****Problem**

Let

$$T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R}), \quad T(M) = M - M^T.$$

- (a) Show that T is a linear operator.
- (b) Consider the following nine matrices in $\mathcal{M}_{3 \times 3}(\mathbb{R})$:

$$\begin{aligned} A &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & B &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & C &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ D &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & E &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, & F &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \\ G &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & H &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, & N &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}. \end{aligned}$$

Let $\mathcal{B} = \{A, B, C, D, E, F, G, H, N\}$. Show that $\mathcal{B}$ is a basis for $\mathcal{M}_{3 \times 3}(\mathbb{R})$.

- (c) Determine the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$.
- (d) Determine a basis for $\text{null}(T)$.
- (e) Determine a basis for $\text{range}(T)$.

Solution**Method 1: Symmetric/skew-symmetric decomposition**

For $M, N \in \mathcal{M}_{3 \times 3}(\mathbb{R})$ and $c \in \mathbb{R}$,

$$T(M + N) = (M + N) - (M + N)^T = (M - M^T) + (N - N^T) = T(M) + T(N),$$

and

$$T(cM) = cM - (cM)^T = c(M - M^T) = cT(M).$$

Thus T is linear.

The matrices A, B, C are the diagonal basis matrices. The matrices D, E, F are symmetric off-diagonal matrices, and G, H, N are skew-symmetric off-diagonal matrices. Since there are nine matrices and

$$\dim \mathcal{M}_{3 \times 3}(\mathbb{R}) = 9,$$

it suffices to prove linear independence.

Suppose

$$\alpha_1 A + \alpha_2 B + \alpha_3 C + \beta_1 D + \beta_2 E + \beta_3 F + \gamma_1 G + \gamma_2 H + \gamma_3 N = 0.$$

Comparing diagonal entries gives

$$\alpha_1 = \alpha_2 = \alpha_3 = 0.$$

Comparing the $(1, 2)$ and $(2, 1)$ entries gives

$$\beta_1 + \gamma_1 = 0, \quad \beta_1 - \gamma_1 = 0,$$

so $\beta_1 = \gamma_1 = 0$. Similarly, comparing the $(1, 3), (3, 1)$ entries gives $\beta_2 = \gamma_2 = 0$, and comparing the $(2, 3), (3, 2)$ entries gives $\beta_3 = \gamma_3 = 0$. Hence all coefficients are zero, and

$$\boxed{\mathcal{B} \text{ is a basis for } \mathcal{M}_{3 \times 3}(\mathbb{R}).}$$

Now A, B, C, D, E, F are symmetric, so T sends each of them to 0. Also G, H, N are skew-symmetric, so $G^T = -G$, $H^T = -H$, and $N^T = -N$. Hence

$$T(G) = 2G, \quad T(H) = 2H, \quad T(N) = 2N.$$

Therefore, in the ordered basis $(A, B, C, D, E, F, G, H, N)$,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}.$$

Since $T(M) = 0$ iff $M = M^T$,

$$\text{null}(T) = \{\text{symmetric } 3 \times 3 \text{ matrices}\}.$$

Thus

$$\boxed{\text{null}(T) = \text{span}\{A, B, C, D, E, F\}.}$$

Also $T(M) = M - M^T$ is always skew-symmetric, and every skew-symmetric K satisfies

$$T\left(\frac{1}{2}K\right) = K.$$

Therefore

$$\boxed{\text{range}(T) = \text{span}\{G, H, N\}.}$$

Method 2: Coordinate action on matrix entries

Writing $M = (m_{ij})$, the (i, j) entry of $T(M)$ is $m_{ij} - m_{ji}$. Thus all diagonal entries of $T(M)$ are zero, and the off-diagonal entries occur in opposite pairs. This immediately shows that the range is the skew-symmetric subspace. The equation $T(M) = 0$ is exactly $m_{ij} = m_{ji}$ for all i, j , which is the symmetric subspace. The given basis $\mathcal{B}$ is arranged precisely into diagonal, symmetric off-diagonal, and skew-symmetric off-diagonal directions, which gives the same matrix representation.

Mark Scheme (20 marks)

• (a) Linearity (3 marks)

- (2) Correct additivity proof.
- (1) Correct homogeneity proof.

- **(b) Basis verification (5 marks)**

- (1) Notes that there are nine matrices and $\dim \mathcal{M}_{3 \times 3}(\mathbb{R}) = 9$.
- (3) Proves linear independence by comparing entries or by decomposing the matrices into standard structural directions.
- (1) Concludes that $\mathcal{B}$ is a basis.

- **(c) Matrix representation (5 marks)**

- (2) Correctly identifies that symmetric basis vectors map to 0.
- (2) Correctly identifies that skew-symmetric basis vectors map to twice themselves.
- (1) Writes the correct 9×9 matrix in the stated order.

- **(d) Nullspace (3 marks)**

- (2) Identifies $\text{null}(T)$ as the symmetric subspace.
- (1) Gives a valid basis.

- **(e) Range (4 marks)**

- (2) Identifies $\text{range}(T)$ as the skew-symmetric subspace.
- (1) Justifies that every skew-symmetric matrix is attained.
- (1) Gives a valid basis.

Question 2**(20 Marks)****Problem**

Let $a \in \mathbb{R}$. Consider

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

- Write down the characteristic polynomial for A and determine its eigenvalues.
- Find all values of a for which A is not diagonalizable.
- For all a for which A is diagonalizable, find a diagonal matrix D and an invertible matrix P such that $A = PDP^{-1}$.

Solution**Method 1: Triangular matrix and eigenspaces**

Since A is upper triangular,

$$\det(A - \lambda I) = (1 - \lambda)(a - \lambda)(2 - \lambda).$$

Thus the eigenvalues are

$$\boxed{1, \quad a, \quad 2.}$$

If $a \neq 1, 2$, then the three eigenvalues are distinct, so A is diagonalizable.

If $a = 1$, then $\lambda = 1$ has algebraic multiplicity 2. We compute

$$A - I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

This matrix has rank 2, so

$$\dim E_1 = 3 - 2 = 1.$$

Since the eigenspace dimension is less than the algebraic multiplicity, A is not diagonalizable.

If $a = 2$, then $\lambda = 2$ has algebraic multiplicity 2. We compute

$$A - 2I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Again the rank is 2, so

$$\dim E_2 = 1,$$

and A is not diagonalizable.

Therefore

$$\boxed{A \text{ is not diagonalizable exactly when } a = 1 \text{ or } a = 2.}$$

Now assume $a \neq 1, 2$. For $\lambda = 1$,

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

For $\lambda = a$,

$$(A - aI)x = 0 \implies x_2 = (a - 1)x_1, \quad x_3 = 0,$$

so take

$$v_2 = \begin{bmatrix} 1 \\ a-1 \\ 0 \end{bmatrix}.$$

For $\lambda = 2$,

$$(A - 2I)x = 0 \implies x_2 = x_1, \quad x_3 = (2-a)x_2,$$

so take

$$v_3 = \begin{bmatrix} 1 \\ 1 \\ 2-a \end{bmatrix}.$$

Therefore

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & a-1 & 1 \\ 0 & 0 & 2-a \end{bmatrix}, \quad D = \text{diag}(1, a, 2), \quad A = PDP^{-1}.$$

Since $a \neq 1, 2$, the triangular matrix P has nonzero diagonal entries and is invertible.

Method 2: Jordan-chain obstruction

The only possible failures of diagonalizability occur when eigenvalues repeat, i.e. $a = 1$ or $a = 2$. In each case, the superdiagonal 1 linking the repeated eigenvalue to an adjacent coordinate creates only one eigenvector for the repeated eigenvalue. Therefore both repeated cases are defective, while all non-repeated cases are diagonalizable. This gives the same conclusion as the direct eigenspace computation.

Mark Scheme (20 marks)

- (a) **Characteristic polynomial and eigenvalues (4 marks)**
 - (2) Uses triangular form to write the characteristic polynomial.
 - (2) States the eigenvalues correctly.
- (b) **Non-diagonalizable values (8 marks)**
 - (2) Notes that only $a = 1$ or $a = 2$ can cause repeated eigenvalues.
 - (2) Shows $a = 1$ is defective.
 - (2) Shows $a = 2$ is defective.
 - (2) Concludes exactly $a = 1, 2$ are not diagonalizable.
- (c) **Diagonalisation for $a \neq 1, 2$ (8 marks)**
 - (3) Finds correct eigenvectors for the three eigenvalues.
 - (2) Forms the correct P and D .
 - (1) Justifies that P is invertible.
 - (2) States $A = PDP^{-1}$ clearly.

Question 3**(20 Marks)****Problem**

Consider the vector space $\mathcal{P}_3(\mathbb{R})$ and the polynomials

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6}, \quad q_1(x) = \frac{x(x-2)(x-3)}{2}, \quad q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

Throughout this question, use

$$\langle f(x), g(x) \rangle = f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

You may use the fact that if two polynomials in $\mathcal{P}_3(\mathbb{R})$ evaluate to the same values at four distinct points, then they are the same polynomial.

- Verify that the product function is a real inner product on $\mathcal{P}_3(\mathbb{R})$.
- Show that $\{q_0(x), q_1(x), q_2(x)\}$ is orthonormal with respect to the inner product.
- Let $f(x) = x^3$. By evaluating at $x = 3$, explain why $f(x)$ does not belong to $\text{span}\{q_0, q_1, q_2\}$. Then explain why $\{q_0, q_1, q_2, f\}$ is a basis for $\mathcal{P}_3(\mathbb{R})$.
- Apply Gram–Schmidt to $\{q_0, q_1, q_2, f\}$ to find q_3 such that $\mathcal{B} = \{q_0, q_1, q_2, q_3\}$ is an orthonormal basis for $\mathcal{P}_3(\mathbb{R})$. Verify that

$$q_3(x) = \frac{x(x-1)(x-2)}{6}.$$

- For a positive integer N , define

$$S(N) = 0^2 + 1^2 + \cdots + N^2.$$

Given that $S(N)$ is a polynomial in N of degree exactly three, write $S(N)$ as a linear combination of polynomials in $\mathcal{B}$.

Solution**Method 1: Evaluation-vector interpretation**

The inner product is the usual dot product of evaluation vectors

$$f \mapsto (f(0), f(1), f(2), f(3)).$$

Linearity in the first slot follows from pointwise addition and scalar multiplication. Symmetry follows from commutativity of real multiplication. Also,

$$\langle f, f \rangle = f(0)^2 + f(1)^2 + f(2)^2 + f(3)^2 \geq 0.$$

If $\langle f, f \rangle = 0$, then $f(0) = f(1) = f(2) = f(3) = 0$. Since $f \in \mathcal{P}_3(\mathbb{R})$ has four distinct roots, it must be the zero polynomial. Hence this is a real inner product.

The polynomials are Lagrange polynomials at the points 0, 1, 2, 3:

$$(q_0(0), q_0(1), q_0(2), q_0(3)) = (1, 0, 0, 0),$$

$$(q_1(0), q_1(1), q_1(2), q_1(3)) = (0, 1, 0, 0),$$

$$(q_2(0), q_2(1), q_2(2), q_2(3)) = (0, 0, 1, 0).$$

Therefore their inner products are exactly dot products of the first three standard basis vectors in $\mathbb{R}^4$, so

$$\langle q_i, q_j \rangle = \delta_{ij} \quad (0 \leq i, j \leq 2).$$

Thus $\{q_0, q_1, q_2\}$ is orthonormal.

Let $f(x) = x^3$. Since

$$f(3) = 27, \quad q_0(3) = q_1(3) = q_2(3) = 0,$$

every polynomial in $\text{span}\{q_0, q_1, q_2\}$ evaluates to 0 at $x = 3$, while $f(3) = 27$. Hence $f \notin \text{span}\{q_0, q_1, q_2\}$. Therefore $\{q_0, q_1, q_2, f\}$ is linearly independent. Since $\dim \mathcal{P}_3(\mathbb{R}) = 4$, it is a basis.

Because q_0, q_1, q_2 are already orthonormal, Gram–Schmidt gives

$$g = f - \langle f, q_0 \rangle q_0 - \langle f, q_1 \rangle q_1 - \langle f, q_2 \rangle q_2.$$

Now

$$\langle f, q_0 \rangle = f(0) = 0, \quad \langle f, q_1 \rangle = f(1) = 1, \quad \langle f, q_2 \rangle = f(2) = 8.$$

Thus

$$g = f - q_1 - 8q_2.$$

Its evaluation vector is

$$(g(0), g(1), g(2), g(3)) = (0, 0, 0, 27).$$

The polynomial

$$q_3(x) = \frac{x(x-1)(x-2)}{6}$$

has evaluation vector $(0, 0, 0, 1)$. Hence $g = 27q_3$, so after normalisation the next orthonormal vector is

$$q_3(x) = \frac{x(x-1)(x-2)}{6}.$$

Therefore

$$\mathcal{B} = \{q_0, q_1, q_2, q_3\}$$

is an orthonormal basis.

Finally, since $\mathcal{B}$ is the Lagrange basis at 0, 1, 2, 3, for any cubic polynomial p ,

$$p(N) = p(0)q_0(N) + p(1)q_1(N) + p(2)q_2(N) + p(3)q_3(N).$$

For $S(N) = 0^2 + 1^2 + \dots + N^2$,

$$S(0) = 0, \quad S(1) = 1, \quad S(2) = 5, \quad S(3) = 14.$$

Therefore

$$S(N) = q_1(N) + 5q_2(N) + 14q_3(N).$$

Method 2: Standard basis in disguise

The map $f \mapsto (f(0), f(1), f(2), f(3))$ is an isomorphism from $\mathcal{P}_3(\mathbb{R})$ to $\mathbb{R}^4$. Under this map,

$$q_0 \mapsto e_1, \quad q_1 \mapsto e_2, \quad q_2 \mapsto e_3, \quad q_3 \mapsto e_4.$$

Thus almost all computations reduce to ordinary dot products and coordinates in $\mathbb{R}^4$. In particular, the expansion of S is simply its evaluation vector

$$(S(0), S(1), S(2), S(3)) = (0, 1, 5, 14),$$

so the coefficient of q_i is $S(i)$.

Mark Scheme (20 marks)

- **(a) Inner product verification (4 marks)**
 - (1) Verifies linearity in the first slot.
 - (1) Verifies symmetry.
 - (1) Verifies non-negativity.
 - (1) Proves positive definiteness using the four distinct roots argument.
- **(b) Orthonormality (4 marks)**
 - (2) Correctly evaluates q_0, q_1, q_2 at 0, 1, 2, 3.
 - (2) Concludes $\langle q_i, q_j \rangle = \delta_{ij}$.
- **(c) Basis argument (3 marks)**
 - (1) Uses evaluation at $x = 3$ to show f is not in the span.
 - (1) Concludes linear independence.
 - (1) Uses $\dim \mathcal{P}_3(\mathbb{R}) = 4$ to conclude basis.
- **(d) Gram–Schmidt and q_3 (5 marks)**
 - (2) Correctly subtracts projections onto q_0, q_1, q_2 .
 - (2) Identifies or verifies $q_3(x) = x(x-1)(x-2)/6$.
 - (1) States that $\mathcal{B}$ is orthonormal.
- **(e) Expansion of $S(N)$ (4 marks)**
 - (2) Computes $S(0), S(1), S(2), S(3)$ correctly.
 - (2) Writes $S(N) = q_1(N) + 5q_2(N) + 14q_3(N)$.

Question 4**(20 Marks)****Problem**

Let J be the $n \times n$ matrix that comprises all ones:

$$J = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}.$$

- Let $\mathbf{j} = (1, 1, \dots, 1)^T$. Explain why $\mathbf{j}$ is an eigenvector of J and determine the corresponding eigenvalue.
- Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $T(x) = Jx$.
 - Explain why T is a linear operator.
 - Determine a basis for $\text{null}(T)$.
- Diagonalize J .
- Let A be the $n \times n$ matrix with $A_{ij} = 0$ if $i = j$ and $A_{ij} = 1$ otherwise. Diagonalize A .

Solution**Method 1: Sum-of-coordinates argument**

Each entry of $J\mathbf{j}$ is the sum of the n entries of $\mathbf{j}$, so

$$J\mathbf{j} = n\mathbf{j}.$$

Hence

$$\boxed{\mathbf{j} \text{ is an eigenvector of } J \text{ with eigenvalue } n.}$$

The map $T(x) = Jx$ is linear because matrix multiplication is linear:

$$T(x + y) = Jx + Jy, \quad T(cx) = cJx.$$

For $x = (x_1, \dots, x_n)^T$,

$$Jx = (x_1 + \cdots + x_n)\mathbf{j}.$$

Thus

$$\text{null}(T) = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = 0\}.$$

A convenient basis is

$$\boxed{\{e_1 - e_n, e_2 - e_n, \dots, e_{n-1} - e_n\}}.$$

The vector $\mathbf{j}$ is an eigenvector with eigenvalue n , and every vector in $\text{null}(T)$ is an eigenvector with eigenvalue 0. Therefore let

$$P = [\mathbf{j} \quad e_1 - e_n \quad e_2 - e_n \quad \cdots \quad e_{n-1} - e_n].$$

The columns form a basis of $\mathbb{R}^n$, so P is invertible. With

$$D_J = \text{diag}(n, 0, 0, \dots, 0),$$

we have

$$J = PD_J P^{-1}.$$

The off-diagonal-ones matrix satisfies

$$A = J - I_n.$$

Thus A has the same eigenvectors as J , with all eigenvalues shifted down by 1. Hence

$$A\mathbf{j} = (n-1)\mathbf{j}, \quad Ax = -x \quad \text{for all } x \in \text{null}(T).$$

Using the same P ,

$$D_A = \text{diag}(n-1, -1, -1, \dots, -1),$$

and

$$A = PD_A P^{-1}.$$

Method 2: Rank-one matrix viewpoint

Let $\mathbf{j} = (1, \dots, 1)^T$. Then

$$J = \mathbf{j}\mathbf{j}^T.$$

Thus

$$Jx = \mathbf{j}(\mathbf{j}^T x),$$

so the range is $\text{span}\{\mathbf{j}\}$ and the nullspace is the hyperplane $\mathbf{j}^T x = 0$. This immediately gives eigenvalue n on $\text{span}\{\mathbf{j}\}$ and eigenvalue 0 on the orthogonal complement. The diagonalisation of $A = J - I_n$ follows by shifting these eigenvalues.

Mark Scheme (20 marks)

• (a) Eigenvector of J (3 marks)

- (2) Computes $J\mathbf{j} = n\mathbf{j}$.
- (1) States the eigenvalue n .

• (b) Linear map and nullspace (6 marks)

- (2) Proves T is linear.
- (2) Identifies $\text{null}(T) = \{x : \sum_i x_i = 0\}$.
- (2) Gives a valid basis for $\text{null}(T)$.

• (c) Diagonalising J (6 marks)

- (2) Uses $\mathbf{j}$ and a nullspace basis as eigenvectors.
- (2) Forms a valid invertible P .
- (2) Gives $D_J = \text{diag}(n, 0, \dots, 0)$ and states $J = PD_J P^{-1}$.

• (d) Diagonalising A (5 marks)

- (2) Recognises $A = J - I_n$.
- (2) Finds eigenvalues $n-1$ and -1 with the same eigenvectors.
- (1) States the diagonalisation $A = PD_A P^{-1}$.

Question 5**(20 Marks)****Problem**

Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces.

- (a) Show that there exists a surjective linear map from $\mathcal{V}$ to $\mathcal{W}$ if and only if

$$\dim(\mathcal{V}) \geq \dim(\mathcal{W}).$$

- (b) Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Show that there exists a linear map T from $\mathcal{V}$ to $\mathcal{W}$ with $\text{null}(T) = \mathcal{U}$ if and only if

$$\dim(\mathcal{U}) \geq \dim(\mathcal{V}) - \dim(\mathcal{W}).$$

Solution**Method 1: Basis construction and rank–nullity**

- (a) Suppose first that $T : \mathcal{V} \rightarrow \mathcal{W}$ is surjective. Then

$$\text{range}(T) = \mathcal{W},$$

so

$$\dim \mathcal{W} = \text{rank}(T) \leq \dim \mathcal{V}.$$

Hence $\dim \mathcal{V} \geq \dim \mathcal{W}$.

Conversely, suppose $\dim \mathcal{V} = n$ and $\dim \mathcal{W} = m$ with $n \geq m$. Choose a basis $v_1, \dots, v_n$ for $\mathcal{V}$ and a basis $w_1, \dots, w_m$ for $\mathcal{W}$. Define

$$T(v_i) = w_i \quad (1 \leq i \leq m), \quad T(v_i) = 0 \quad (m < i \leq n),$$

and extend linearly. The range contains $w_1, \dots, w_m$, so T is surjective. Therefore

$$\boxed{\exists \text{ surjective } T : \mathcal{V} \rightarrow \mathcal{W} \iff \dim \mathcal{V} \geq \dim \mathcal{W}.}$$

- (b) Suppose $T : \mathcal{V} \rightarrow \mathcal{W}$ is linear with $\text{null}(T) = \mathcal{U}$. By rank–nullity,

$$\dim \mathcal{V} = \dim \text{null}(T) + \text{rank}(T) = \dim \mathcal{U} + \text{rank}(T).$$

Since $\text{rank}(T) \leq \dim \mathcal{W}$,

$$\dim \mathcal{V} \leq \dim \mathcal{U} + \dim \mathcal{W},$$

so

$$\dim \mathcal{U} \geq \dim \mathcal{V} - \dim \mathcal{W}.$$

Conversely, suppose

$$\dim \mathcal{U} \geq \dim \mathcal{V} - \dim \mathcal{W}.$$

Let

$$k = \dim \mathcal{U}, \quad n = \dim \mathcal{V}, \quad m = \dim \mathcal{W}.$$

Then $n - k \leq m$. Choose a basis $u_1, \dots, u_k$ for $\mathcal{U}$ and extend it to a basis of $\mathcal{V}$:

$$u_1, \dots, u_k, v_1, \dots, v_{n-k}.$$

Since $n - k \leq m$, choose linearly independent vectors $w_1, \dots, w_{n-k}$ in $\mathcal{W}$. Define $T : \mathcal{V} \rightarrow \mathcal{W}$ on this basis by

$$T(u_i) = 0 \quad (1 \leq i \leq k), \quad T(v_j) = w_j \quad (1 \leq j \leq n - k),$$

and extend linearly. Then $\mathcal{U} \subseteq \text{null}(T)$. Conversely, if

$$x = \sum_{i=1}^k \alpha_i u_i + \sum_{j=1}^{n-k} \beta_j v_j$$

and $T(x) = 0$, then

$$0 = T(x) = \sum_{j=1}^{n-k} \beta_j w_j.$$

Since the w_j are linearly independent, all $\beta_j = 0$, so $x \in \mathcal{U}$. Hence

$$\text{null}(T) = \mathcal{U}.$$

Therefore

$$\boxed{\exists T : \mathcal{V} \rightarrow \mathcal{W} \text{ with } \text{null}(T) = \mathcal{U} \iff \dim \mathcal{U} \geq \dim \mathcal{V} - \dim \mathcal{W}.}$$

Method 2: Quotient-space viewpoint

For part (b), a linear map $T : \mathcal{V} \rightarrow \mathcal{W}$ with $\text{null}(T) = \mathcal{U}$ is equivalent to an injective linear map

$$\bar{T} : \mathcal{V}/\mathcal{U} \rightarrow \mathcal{W}.$$

Such an injective map exists exactly when

$$\dim(\mathcal{V}/\mathcal{U}) \leq \dim \mathcal{W}.$$

Since

$$\dim(\mathcal{V}/\mathcal{U}) = \dim \mathcal{V} - \dim \mathcal{U},$$

this condition is exactly

$$\dim \mathcal{U} \geq \dim \mathcal{V} - \dim \mathcal{W}.$$

This is a compact alternative to the explicit basis construction.

Mark Scheme (20 marks)

• (a) Surjective map criterion (8 marks)

- (3) Proves necessity using $\text{rank}(T) = \dim \mathcal{W} \leq \dim \mathcal{V}$.
- (3) Constructs a linear map from a basis of $\mathcal{V}$ onto a basis of $\mathcal{W}$ when $\dim \mathcal{V} \geq \dim \mathcal{W}$.
- (1) Justifies that the constructed map is linear.
- (1) Concludes surjectivity.

• (b) Prescribed nullspace criterion (12 marks)

- (3) Proves necessity using rank-nullity and $\text{rank}(T) \leq \dim \mathcal{W}$.
- (2) Sets $k = \dim \mathcal{U}$, $n = \dim \mathcal{V}$, $m = \dim \mathcal{W}$ and obtains $n - k \leq m$.
- (2) Extends a basis of $\mathcal{U}$ to a basis of $\mathcal{V}$.
- (2) Chooses enough linearly independent vectors in $\mathcal{W}$ and defines T on the basis.
- (2) Proves that the nullspace of the constructed map is exactly $\mathcal{U}$.
- (1) States the final iff conclusion.